

Machine Learning and its Applications

Today's goals



Module 1

Machine Learning Fundamentals I: Basic Concepts

Welcome

Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations

Video • 6 min

Specialization and Course Structure

Reading • 10 min

Specialization Sample Certificate

Reading • 10 min

Stay connected with UofG Online

Reading • 5 min

Module 7 Introduction

Introduction to Module 7's Study

Video • 1 min

How to Access MATLAB Online

Reading • 5 min

Machine Learning in Engineering Practice: Tool, Replacement, or Decision Aid?

Reading • 5 min

Module 7-1 Machine Learning Fundamentals

Machine Learning Fundamentals: What is Machine Learning

Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (1)

Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (2)

Video • 8 min

Materials for Machine Learning Fundamentals

Reading • 10 min

From Explicit Rules to Learned Models: What Actually Changes?

Reading • 5 min

Module 1 Learning Coach

Dialogue • 5 min

Module 7-2 Mapping Inputs to Outputs

Mapping Inputs to Outputs: Data Representation

Video • 10 min

Mapping Inputs to Outputs: Parametric ML Model

Video • 7 min

Mapping Inputs to Outputs: Non-Parametric ML Model

Video • 7 min

Mapping Inputs to Outputs: Evaluate Output

Video • 2 min

Materials for Mapping Inputs to Outputs

Reading • 10 min

How Data Representation and Model Choice Shape What a Model Can Learn

Reading • 5 min

Module 7-3 MATLAB Implementation

MATLAB Implementation: Simple Linear Regression and KNN

Video • 6 min

Materials about MATLAB Implementation

Reading • 10 min

From Concept to Code: What the Model Is Actually Doing

Reading • 5 min

Module 7 Quiz

Module 7 Quiz

Practice Assignment • 30 min

Module 7 Assignments

Module 7 Assignment 1

Graded App Item • 30 min

Module 7 Assignment 2

Graded App Item • 30 min

CSV File for Module 7 Assignment

Reading • 1 min

Module 7 Recap

Reading • 5 min

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 3

Data Preparation

Module 4

Support Vector Machines

Module 5

Artificial Neural Networks

Machine Learning in Engineering Practice: Tool, Replacement, or Decision Aid?



Machine learning is widely used in science and engineering for **classification and regression**, but its role is often misunderstood or overstated.

Identify **one concrete engineering or scientific task** where machine learning is genuinely useful today. Be specific about:

- what the input data looks like,
- what the output represents, and
- whether the task is classification or regression.

Is machine learning in your example:

- replacing a traditional analytical or rule-based method,
- complementing it, or
- mainly acting as a decision-support tool? Explain why, not just what.

From this module's introduction, which **core concepts** (data, model, training, prediction, evaluation) do you think are most critical for **trusting** machine-learning results in engineering or science, and which ones are commonly misunderstood?

Go to next item

✓ Completed

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